

PARTH SARTHI SAXENA

PROFESSIONAL SUMMARY Analytical engineering student with experience in quantitative trading, strategy modeling, and data-driven risk evaluation. Skilled in backtesting, performance optimization, and applying probabilistic methods to assess expected value and risk-adjusted returns. Interested in quantitative trading, financial modeling, and systematic strategy development.

EDUCATION JECRC Foundation , Jaipur
B.Tech : Electronics and communication engineering (may 2027)
CGPA : 8.21
CBSE 10TH :85% (Bakshies Springdales)
CBSE 12TH :75% (Lord Buddha)

TECHNICAL PROJECTS

Trading Strategy Backtesting Engine (Python)

- Designed and implemented a backtesting framework to evaluate trading strategies using historical time-series data.
- Computed performance metrics including win rate, maximum drawdown, risk-reward ratio, and risk-adjusted returns.
- Applied expectancy (expected value) calculations to assess long-term profitability of strategies.
- Simulated strategy performance across varying volatility and liquidity regimes.
- Utilized pandas and numpy for structured data processing and quantitative analysis.
- Optimized strategy parameters through historical data simulation and performance comparison.
- Designed modular backtesting architecture enabling easy integration of multiple trading strategies.

INDEPENDENT TRADING & QUANTITATIVE ANALYSIS

- Designed and implemented systematic trading strategies evaluated over 400+ trades, achieving consistent win rates (51%–56%) with optimized risk-reward ratios (1:3–1:5) and controlled drawdown (~10–12%).
- Modeled strategy performance under different market regimes, analyzing the impact of volatility and liquidity on execution efficiency.
- Applied probabilistic reasoning and expected value concepts to validate trade selection and long-term profitability.
- Developed rule-based trading systems and evaluated performance using structured journaling and data-driven analysis.
- Analyzed correlation between volume expansion and short-term momentum to identify statistically favorable entry conditions.
- Studied effects of leverage, slippage, and behavioral biases on trading outcomes in multi-asset environments.

PROJECTS	Financial analysis of Interglobe aviation - Financial Management	April 2025 - May 2025
	• Performed financial modeling and valuation analysis, evaluating profitability, IPO performance, and earnings impact. • Analyzed market reaction to financial disclosures and identified key drivers of price movement. • Assessed capital structure, institutional ownership, and valuation gaps influencing stock behavior.	
	Market Analysis – Sundaram Clayton	Dec 2025-Jan 2026
	• Conducted quantitative ratio analysis including liquidity, solvency, leverage, and activity metrics. • Benchmarked financial performance against industry peers to evaluate investment potential. • Modeled historical price behavior to identify key trends and influencing factors.	

SKILLS

Languages: Python, C++

Core: Data Structures & Algorithms, Probability, Statistics

Quantitative: Expected Value, Risk Modeling, Time Series Analysis

Tools: Python (pandas, numpy), Excel
